MAT 2000: Introduction to Statistics
Grace College
Quiz 1.1-3, Form: A

Name : _____

Match the title of the six-step method with the descriptions.

_____ 1. Ask a research question.

_____ 2. Design a study and collect data.

_____ 4. Draw inferences beyond the data.

_____ 5. Formulate conclusions.

_____ 3. Explore the data.

_____ 6. Look back and ahead.

(a) Point out limitations of the study and suggest new studies to build on the findings.

(b) Look for patterns that relate to the research question or raise additional questions.

(c) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(d) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(e) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(f) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

Match each term from section 1.1 with it's definition

_____ observational unit

_____ quantitative variable

_____ variable

_____ data

_____ categorical variable

_____ distribution

(a) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

(b) the individual object on which data are recorded

(c) the pattern of outcomes of a variable

(d) a characteristic being recorded about the observational units

(e) places observational units into categories

(f) the values measured or categories recorded on individual objects of interest

Match each term from section 1.2 with it's definition

_____ Unusual observations

_____ histogram

_____ dotplot

_____ variability of a distribution

_____ shape of a distribution

_____ center of a distribution

(a) the middle or typical value

(b) the spread in the observations

(c) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(d) a graph with one dot representing the variable outcome for each observational unit

(e) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(f) observations that do not follow the overall pattern of the distribution

Match each term from section 1.3 with it's definition

_____ probability of an outcome

_____ random process

_____ simulation

(a) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run

(b) an artificial re-creation of a random process

(c) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

Answer Key for Exam A

Match the title of the six-step method with the descriptions.

(a) 1. Ask a research question.

(b) 2. Design a study and collect data.

(a) 4. Draw inferences beyond the data.

(b) 5. Formulate conclusions.

(c) 3. Explore the data.

(f) 6. Look back and ahead.

(a) Point out limitations of the study and suggest new studies to build on the findings.

(b) Look for patterns that relate to the research question or raise additional questions.

(c) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(d) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(e) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(f) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

Match each term from section 1.1 with it's definition

(e) observational unit

(c) quantitative variable

(b) variable

(d) data

(f) categorical variable

(a) distribution

(a) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

(b) the individual object on which data are recorded

(c) the pattern of outcomes of a variable

(d) a characteristic being recorded about the observational units

(e) places observational units into categories

(f) the values measured or categories recorded on individual objects of interest

Match each term from section 1.2 with it's definition

(c) Unusual observations

(b) histogram

(f) dotplot

(e) variability of a distribution

(d) shape of a distribution

(a) center of a distribution

(a) the middle or typical value

(b) the spread in the observations

(c) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(d) a graph with one dot representing the variable outcome for each observational unit

(e) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(f) observations that do not follow the overall pattern of the distribution

Match each term from section 1.3 with it's definition

(e) probability of an outcome

(d) random process

(c) simulation

(a) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run

(b) an artificial re-creation of a random process

(c) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

MAT 2000: Introduction to Statistics
Grace College
Quiz 1.1-3, Form: B

Name : _____

Match the title of the six-step method with the descriptions.

_____ 3. Explore the data.

_____ 1. Ask a research question.

_____ 4. Draw inferences beyond the data.

_____ 2. Design a study and collect data.

_____ 6. Look back and ahead.

_____ 5. Formulate conclusions.

(a) Look for patterns that relate to the research question or raise additional questions.

(b) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(c) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(d) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(e) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

(f) Point out limitations of the study and suggest new studies to build on the findings.

Match each term from section 1.1 with it's definition

_____ observational unit

_____ quantitative variable

_____ distribution

_____ categorical variable

_____ variable

_____ data

(a) a characteristic being recorded about the observational units

(b) places observational units into categories

(c) the individual object on which data are recorded

(d) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

(e) the values measured or categories recorded on individual objects of interest

(f) the pattern of outcomes of a variable

Match each term from section 1.2 with it's definition

_____ histogram

_____ center of a distribution

_____ Unusual observations

_____ dotplot

_____ shape of a distribution

_____ variability of a distribution

(a) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(b) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(c) the middle or typical value

(d) a graph with one dot representing the variable outcome for each observational unit

(e) the spread in the observations

(f) observations that do not follow the overall pattern of the distribution

Match each term from section 1.3 with it's definition

_____ simulation

_____ random process

_____ probability of an outcome

(a) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

(b) an artificial re-creation of a random process

(c) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run

Answer Key for Exam B

Match the title of the six-step method with the descriptions.

(a) 3. Explore the data.

(c) 1. Ask a research question.

(c) 4. Draw inferences beyond the data.

(b) 2. Design a study and collect data.

(f) 6. Look back and ahead.

(e) 5. Formulate conclusions.

(a) Look for patterns that relate to the research question or raise additional questions.

(b) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(c) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(d) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(e) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

(f) Point out limitations of the study and suggest new studies to build on the findings.

Match each term from section 1.1 with it's definition

(d) observational unit

(b) quantitative variable

(f) distribution

(e) categorical variable

(a) variable

(c) data

(a) a characteristic being recorded about the observational units

(b) places observational units into categories

(c) the individual object on which data are recorded

(d) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

(e) the values measured or categories recorded on individual objects of interest

(f) the pattern of outcomes of a variable

Match each term from section 1.2 with it's definition

(c) histogram

(d) center of a distribution

(f) Unusual observations

(e) dotplot

(a) shape of a distribution

(b) variability of a distribution

(a) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(b) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(c) the middle or typical value

(d) a graph with one dot representing the variable outcome for each observational unit

(e) the spread in the observations

(f) observations that do not follow the overall pattern of the distribution

Match each term from section 1.3 with it's definition

(a) simulation

(d) random process

(b) probability of an outcome

(a) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

(b) an artificial re-creation of a random process

(c) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run

MAT 2000: Introduction to Statistics
Grace College
Quiz 1.1-3, Form: C

Name : _____

Match the title of the six-step method with the descriptions.

_____ 4. Draw inferences beyond the data.

_____ 1. Ask a research question.

_____ 6. Look back and ahead.

_____ 3. Explore the data.

_____ 2. Design a study and collect data.

_____ 5. Formulate conclusions.

(a) Point out limitations of the study and suggest new studies to build on the findings.

(b) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(c) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(d) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(e) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

(f) Look for patterns that relate to the research question or raise additional questions.

Match each term from section 1.1 with it's definition

_____ data

_____ distribution

_____ quantitative variable

_____ variable

_____ categorical variable

_____ observational unit

(a) the values measured or categories recorded on individual objects of interest

(b) the individual object on which data are recorded

(c) a characteristic being recorded about the observational units

(d) places observational units into categories

(e) the pattern of outcomes of a variable

(f) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

Match each term from section 1.2 with it's definition

_____ variability of a distribution

_____ histogram

_____ Unusual observations

_____ dotplot

_____ center of a distribution

_____ shape of a distribution

(a) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(b) the middle or typical value

(c) observations that do not follow the overall pattern of the distribution

(d) a graph with one dot representing the variable outcome for each observational unit

(e) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(f) the spread in the observations

Match each term from section 1.3 with it's definition

_____ random process

_____ simulation

_____ probability of an outcome

(a) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

(b) an artificial re-creation of a random process

(c) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run

Answer Key for Exam C

Match the title of the six-step method with the descriptions.

(b) 4. Draw inferences beyond the data.

(c) 1. Ask a research question.

(c) 6. Look back and ahead.

(a) 3. Explore the data.

(b) 2. Design a study and collect data.

(f) 5. Formulate conclusions.

(a) Point out limitations of the study and suggest new studies to build on the findings.

(b) Determine whether any findings in the data reflect a genuine tendency and estimate the strength of that tendency.

(c) Plan how to select the people or objects to be studied, decide how to gather the relevant data, and carry out the data collection in a careful, systematic manner.

(d) Confirm the question can be addressed by collecting data. Typical questions involve comparing groups, asking how two characteristics are related, or surveying people's opinions.

(e) Consider whether inferences made in step 4 can be generalized to processes or groups beyond those represented in the study. Identify whether a cause-and-effect conclusion is warranted.

(f) Look for patterns that relate to the research question or raise additional questions.

Match each term from section 1.1 with it's definition

(d) data

(a) distribution

(b) quantitative variable

(f) variable

(e) categorical variable

(c) observational unit

(a) the values measured or categories recorded on individual objects of interest

(b) the individual object on which data are recorded

(c) a characteristic being recorded about the observational units

(d) places observational units into categories

(e) the pattern of outcomes of a variable

(f) has numeric values for which ordinary arithmetic (addition, multiplication) makes sense

Match each term from section 1.2 with it's definition

(d) variability of a distribution

(b) histogram

(e) Unusual observations

(a) dotplot

(f) center of a distribution

(c) shape of a distribution

(a) described by the number of peaks and whether one side of the distribution tends to stretch out longer than the other

(b) the middle or typical value

(c) observations that do not follow the overall pattern of the distribution

(d) a graph with one dot representing the variable outcome for each observational unit

(e) a graph that separates the quantitative variable into bins and displays the number of observations in each bin

(f) the spread in the observations

Match each term from section 1.3 with it's definition

(d) random process

(e) simulation

(a) probability of an outcome

(a) the long-run proportion of times an outcome from a random process occurs if the process is repeated indefinitely under identical conditions

(b) an artificial re-creation of a random process

(c) a repeatable process where the result for an individual attempt is not known in advance but there is a predictable pattern to the results in the long run